Estimation of $PM_{2.5}$ Concentration in DKI Jakarta from Sentinel-5P Imagery by Considering Meteorological Factors Using Random Forest Approach

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Abstract. Poor air quality, caused by high pollutant levels, harms the environment and public health. Fine particulate matter ($PM_{2.5}$), less than 2.5 μm in diameter, is a major concern in air quality observations and is a major concern due to its ability to penetrate the respiratory system, increasing risks of lung cancer, premature death, and unnatural births. Jakarta faces severe air pollution, yet its air quality monitoring network remains limited. To address this, this study employs machine learning, specifically random forest algorithms, using spatial regression to model $PM_{2.5}$ levels. The variables used are meteorological elements and particulates and gasses obtained by utilizing remote sensing. It was found that the R^2 value of 0.793 implies that the accuracy of the variables used reaches 79.3 percent and the RMSE value of 8.28 μg/m3. The spatial pattern formed in this spatial modelling follows the pattern of the rainy season and dry season, where the highest value of the spatial pattern of the $PM_{2.5}$ parameter is in the JJA month (June, July and August), and finally at the lowest value in the DJF month (December, January and February).

1. Introduction

Environmental pollution has become a serious social problem along with economic growth and accelerated industrialization. [1]. Fossil fuels, biomass, and dust generated during construction are the main causes of environmental pollution. Fine particles smaller than or equal to 2.5 microns fall under this category. $PM_{2.5}$ (Particulate Matter) refers to fine respirable airborne particles with an aerodynamic diameter of 2.5 μm or smaller [2]. Epidemiological studies show that $PM_{2.5}$ concentrations can efficiently penetrate the human respiratory system, increasing the risk of lung cancer, unnatural births, premature death, unnatural births, and premature death [3]. In addition, it results in problems such as acid rain, reduced visibility, ecosystem degradation, and climate change [4].

Air quality monitoring in Jakarta has been conducted continuously by the government or central government and international organizations. Measurement of $PM_{2.5}$ concentration using the Beta Attenuation Monitoring method with units of micrograms per cubic meter (μm/m^3). However, the scattered coverage of air quality monitoring locations has not been able to describe the detailed and spatially coherent mapping of $PM_{2.5}$ concentrations. This limitation makes it a problem for decision-making in designing air quality policies [5]. The ability to accurately map pollutant concentrations and identify areas exposed to air pollution is important for air quality management [6]. Therefore, air pollution modeling and mapping are useful instruments for predicting and estimating air pollutants and assessing their impacts on human health, the environment, and the economy [7].

Given the vast coverage and diversity of PM loadings in the environment, remote sensing data from Earth-orbiting satellites can be used to bridge this gap [8]. Remote Sensing is the science and art of obtaining information about objects, areas, or phenomena through data analysis from a device that does not interact directly with the object, area, or phenomenon [9]. Since the lack of historical data and field monitoring coverage are obstacles to studying the problem, spatial modeling is required. The ability to accurately map pollutant concentrations and identify areas exposed to air pollution is important for air quality management.

In recent years, the application of machine learning algorithms is increasing. Machine learning is able to solve the problem of complex non-linear relationships between observed and predicted variables in the dataset [10]. Many studies have applied various machine learning algorithms for $PM_{2.5}$ estimation. Previous research has utilized satellite and meteorological data as predictor variables for $PM_{2.5}$ estimation in Malaysia by exploring several machine-learning algorithms and producing random forest as the best algorithm [11]. Random forest (RF) is based on decision tree theory, with the basic principle of building each tree based on

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samples and then dividing the samples based on the best randomly selected partition points [12]. Therefore, this research will use the best-performing algorithm in the previous study, Random Forest, to create a spatial-temporal model of air quality in Jakarta.

This research aims to better predict $PM_{2.5}$ values in various areas in Jakarta, to support early warning that will be used to take air pollution control measures. In this study, we used Sentinel-5P image pollutant data and meteorological data as predictors to build a machine-learning algorithm. The dataset is divided into 80% as training data and 20% as testing data. The training dataset is used to train the models, while the testing dataset is used to evaluate the model performance of the trained models. This research uses Google Earth Engine (GEE) as a tool to build a random forest model. For visualization, we used QGIS. Thus, this research is expected to be a problem solver for the limitations of ground-level monitoring networks and an initial study for strategies and policies to mitigate the impact of $PM_{2.5}$ concentrations in Jakarta.

2. Material and Methods

2.1 Study area

Severe pollution caused by fine particulate matter dominated by $PM_{2.5}$ can pose a series of economic, environmental, and health problems. Jakarta, the capital of Indonesia, is often ranked as the country's capital with poor air quality by IQAir, so the study of PM2.5 is very important. This study's focus area is Jakarta, located at coordinates 6°12' - 6°23 LS and 106°48' - 106°58' BT with an area of 661.23 km2. Jakarta is a lowland city with an average height of +7 meters above sea level [13]. The locations of the monitoring stations are shown in **Figure 1.**

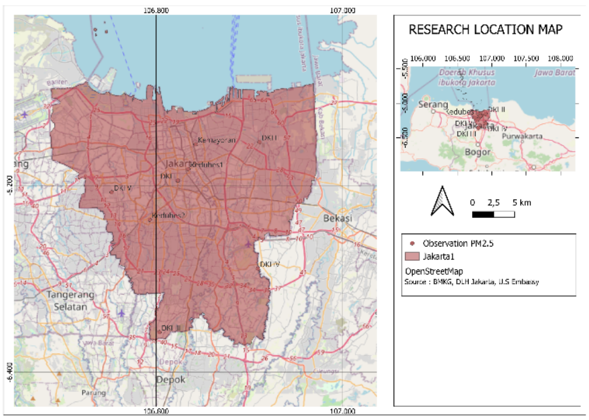


Fig. 1. Study area

In this study, $PM_{2.5}$ monitoring stations managed by the government were selected to obtain daily PM2.5. The coordinates of each observation station are shown in **Table 1**.

Table 1. Ground-level monitoring networks

No	Observation Station	Longitude	Latitude
1	Stasiun Meteorologi Kemayoran (Kemayoran Meteorological Station)	106.842528	-6.155914
2	DKI1	106.8235	-6.19466
3	DKI2	106.91089	-6.15357
4	DKI3	106.8037	-6.35693
5	DKI4	106.9092	-6.28889
6	DKI5	106.75256	-6.207255
7	Kedubes Amerika Serikat (Jakarta Pusat) (U.S. Embassy – Central Jakarta)	106.834489	-6.182458
8	Kedubes Amerika Serikat (Jakarta Selatan) (U.S. Embassy - South Jakarta)	106.793287	-6.155914

2.2 Data Collection

This study uses $PM_{2.5}$ field data as label data. Pollutant gases from Sentinel-5P data and meteorological data as predictor variables in building machine learning models. The period of this research is January 2021 - December 2023, where post-COVID or new normal conditions have been applied, as for the last 3 years, the condition of returning to normal activities and outdoor activities are no longer restricted, and industrial activities are also running optimally.

The data needed for the modeling are available to Google Earth Engine (GEE), except for the field data $PM_{2.5}$, which are gathered from multiple sources in Jakarta: BMKG, the Government of Jakarta, and the Embassy of the United States in Indonesia.

2.2.1 TropOMI Data

This study uses the TropOMI offline version of the L3 product released by the Google Earth Engine (GEE). TropOMI is a tropospheric observer carried on board the Sentinel-5P satellite launched by the European Space Agency (ESA) on 13 October 2017. This satellite is a solar synchronous orbit satellite operating at an altitude of 824 km from the ground, with an operation cycle of 16 days. Since 6 August 2019, TropOMI has had a spatial resolution of 7 km × 3.5 km, improved from the 5.5 km × 3.5 km prior to that date. However, the L3 product offered by Google Earth Engine is a regrid product with a resolution of 1,132.2 meters.

2.2.2 Meteorological Data

The RF machine learning model created in this study uses meteorological data for 2021-2023. The data include temperature at 2 m height, zonal (U), and meridional (V) wind components (shown in Table 2). Among them, temperature, U, and V wind components are derived from daily ERA 5 Land data, with a spatial resolution of 11,132 meters.

Table 2. Research variables

No	Variable	Unit	Resolution
1	PM2.5	$\mu g/m^3$	Data Point
2	AOD (*Aerosol Optical Depth*)	-	1.113,2 meter
3	CO	mol/m^2	1.113,2 meter
4	SO2	mol/m^2	1.113,2 meter
5	NO2	mol/m^2	1.113,2 meter
6	O3	mol/m^2	1.113,2 meter
7	U component of wind	m/s	11.132 meters
8	V component of wind	m/s	11.132 meters

2.2.3 Ground Monitoring Station Data

The ground-monitored PM2.5 concentration data used in this study came from daily data provided by BMKG, Government of Jakarta, and Embassy of the United States in Indonesia. The study period is from January 1, 2021, to December 31, 2023.

2.3 Data Preprocessing

2.3.1 Generated Wind Speed and Direction

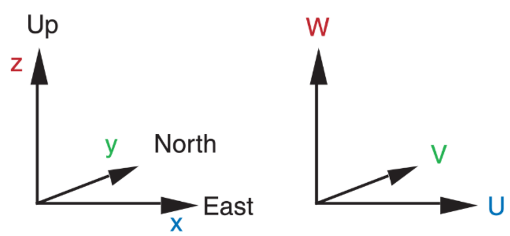


Fig. 2. Cartesian coordinates and velocity components

In meteorology, the U component (zonal wind) is positive for west-to-east flow (east wind), and the V component (meridional wind) is positive for south-to-north flow (north wind). Its relationship to wind direction and speed can be converted to this formula:

Wind speed

$$M = (U^2 + V^2)^{1/2} \tag{1}$$

Wind Direction

$$\alpha = 90° - \frac{360°}{C} \cdot arctan\left(\frac{V}{U}\right) + \alpha_o \tag{2}$$

where α_o = 180° if U > 0, but is zero otherwise. C is the angular rotation in a full circle (C = 360° = 2·π radians). U and v refer to and components of wind [14].

2.3.2 Data Cleaning

Data cleaning refers to the process of identifying, correcting, and removing invalid, empty, or corrupted data on each variable to be used [15]. This process aims to reduce the possibility of bias or error in the learning machine modeling process. In the satellite data cleaning process, the first step is to select data that has a complete data set. Next, the data on each variable is re-examined to identify data corruption and separate the data from the data set. In this case, each dataset allows for missing and unbalanced data values, so it is necessary to do data cleaning and data analysis. In this study, to handle missing data by deleting every row that has an empty value.

This step is important to ensure the quality and completeness of the data before further processing. As a result, raw data was obtained as predictor variables and target variables. In this study, 3.588 raw data were generated, as shown in **Figure 3.** where there were no missing values, outliers, and corrupted data.

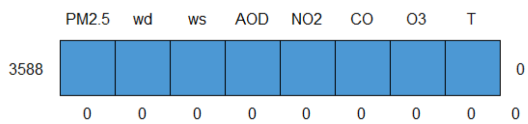


Fig. 3. Amount of the data availability

2.4 Method

In recent years, machine learning algorithms have been widely used in weather inversion. Since machine learning requires a large amount of observation data for training, the first task is to collect a large amount of data and construct a dataset. In this study, the machine learning components used in this research include random forest (RF) (**Fig. 4**). All data sets were divided into a test set and a training set; the model used the data from January 2021 – December 2023 as the testing set and training set.

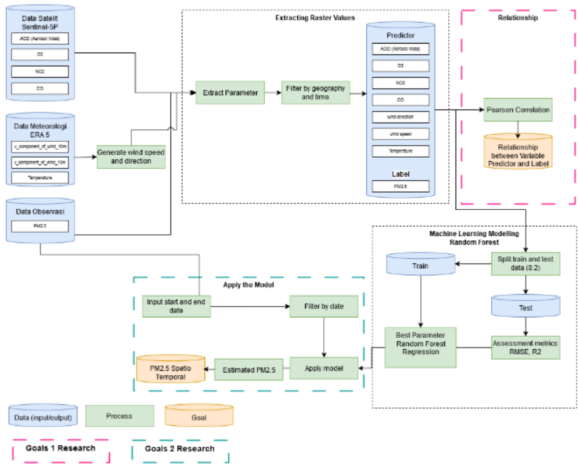


Fig. 4. Flowchart of the proposed method for estimation PM2.5 using the Random Forest (RF) algorithm

This research divides the dataset into 80% training data and 20% test data. This division is based on previous research that obtained the optimal model at 80% : 20%. After that, hyperparameters are used to control the quality of the training process improving the efficiency and accuracy of the model. The evaluation metric can be done with the coefficient of determination (R^2) and RMSE.

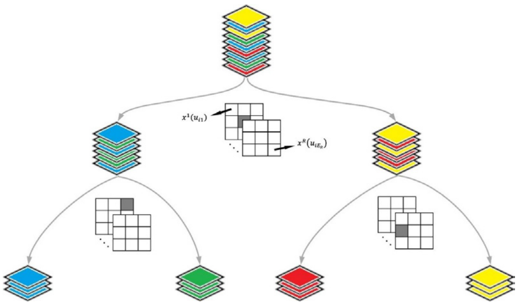


Fig. 5. Spatial Random Forests Algorithm for Geoscience Data Analysis and Modelling
(Source Talebi, 2022)

Random forest is a combination of predictions for a set of trees so that the value of the random vector that is sampled is different for each tree and the same distribution for all samples [12].

2.4.1 Evaluation Procedure

The statistical metrics used in this study to measure the prediction performance of the developed model included: the coefficient of determination (R^2) and RMSE.

$$R^2 = \frac{\sum(\hat{y}-\bar{y})^2}{\sum(y-\bar{y})^2} = 1 - \frac{\sum(y-\hat{y})^2}{\sum(y-\bar{y})^2} \quad (3)$$

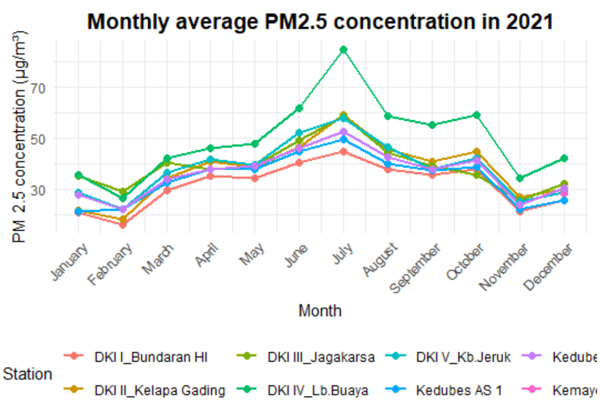
in which $\sum(y - \hat{y})^2$ is the sum of the squares of residuals, i.e., the sum of the squares of the deviation between the data points and their corresponding positions on the regression line, and $\sum(y - \bar{y})^2$ is the total sum of squares, i.e., the sum of the squares of the deviation between the data point and its average value. When R^2 is closer to 1, it indicates that the goodness-of-fit of the model is higher.

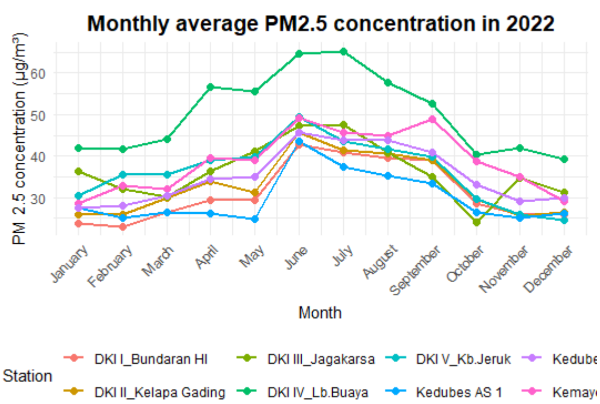
Root Mean Squared Error (RMSE) is an evaluation metric that calculates the square root of the mean square of the difference between the actual and predicted values. It is very sensitive to extremely large or small errors in a set of data. RMSE is often used in regression model evaluation because it has the same units as the target variable, making it easier to interpret. RMSE as shown in Equation (4)

$$RMSE = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(y'_i - y_i)^2} \quad (4)$$

3. Result and Discussion

Figure 6 shows the pattern of $PM_{2.5}$ concentrations at all observation points. It can be seen that $PM_{2.5}$ concentrations fluctuate every year. The pattern shows an increase in $PM_{2.5}$ concentration values during the dry season in JJA (June - July - August) and tends to decrease during the rainy season in DJF (December - January - February). This shows the pattern of $PM_{2.5}$ concentrations in accordance with seasonal patterns in Indonesia.





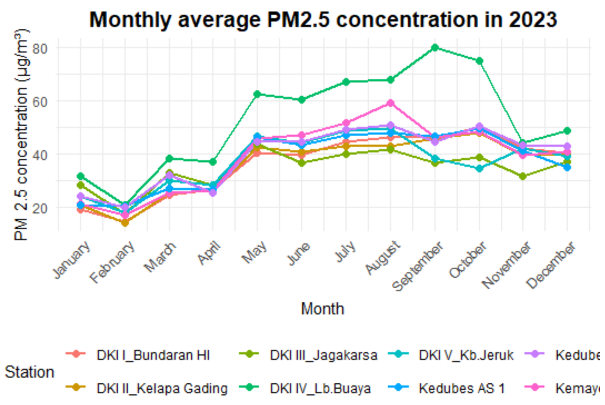


Fig. 6. Monthly Average PM2.5 Concentration in Jakarta (2021-2023)

After collecting the entire dataset, we divided it into training and testing data, with an 80% allocation for training and 20% for testing. In spatial modeling, the main predictor is required as a material or spatial-temporal modeling. Long-range data with spatial data can be used as spatial modelling. The model formed can be influenced by the method taken. The selection and use of the random forest method provides a good reference in predicting $PM_{2.5}$ parameter values. Random forest spatial regression yields accurate predictions, especially when paired with dense observational data (**Fig. 7**). A correlation value of 0.793 was obtained in the modeling conducted between the predicted results and the observed values, indicating a strong relationship. Additionally, the model achieved an RMSE value of 7.83 µg/m³, which further demonstrates its reliability. The R^2 value of 0.793 suggests that the accuracy of the variables used in the model reaches 79.3%, highlighting the effectiveness of the random forest approach in this context.

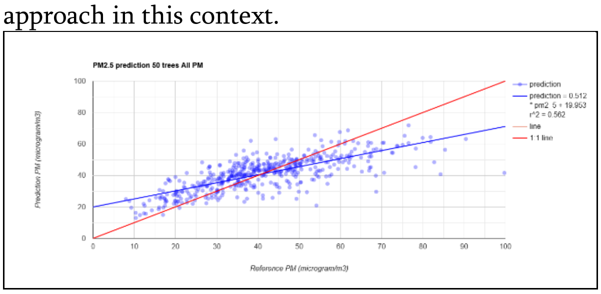


Fig. 7. Random forest model

The RF model has the independence of multiple modeling and the randomness of variable selection. Moreover, it can compare the accuracy change degree of specific independent variables when they participate in modelling and when they do not participate in modelling. Therefore, it can reflect the importance score of each independent variable in regression.

The correlation between explanatory variables and dependent variable, i.e., the $PM_{2.5}$ concentration at the ground site, was calculated and the results are shown in **Figure 5.** It can be seen that the correlation between the $PM_{2.5}$ concentration and wind speed is the good negative correlation, at -0.38. It can also be seen from **Figure 5.** that the correlation between explanatory variables is not high, and some variables have almost no relationship with each other. These results suggest that these explanatory variables meet the conditions of mutual independence and can be selected for model regression analysis.

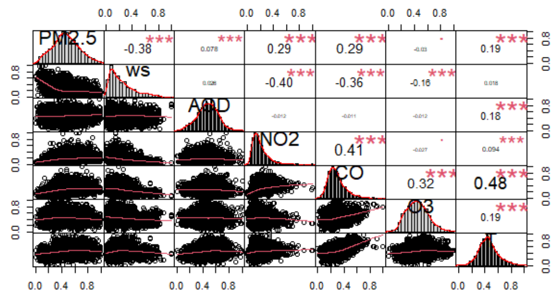


Fig. 8. Relative importance of explanatory variables to the model

Figure 8 shows the visualization of the best random forest model processed using QGIS. It appears that by using the random forest model, and QGIS visualization we can find out the distribution of $PM_{2.5}$ values in the Jakarta area. The author hopes that this can later be used as a problem solver for air quality management in Jakarta.

In the DJF month, PM2.5 concentrations tend to be low; this is due to the high amount of rainfall in that month. However, in the JJA month PM2.5 concentrations tend to be high, due to the low amount of rainfall. Based on the relative importance value in the model, the wind speed variable shows the highest rank. This can be strengthened by the distribution of wind speed in the visualization.

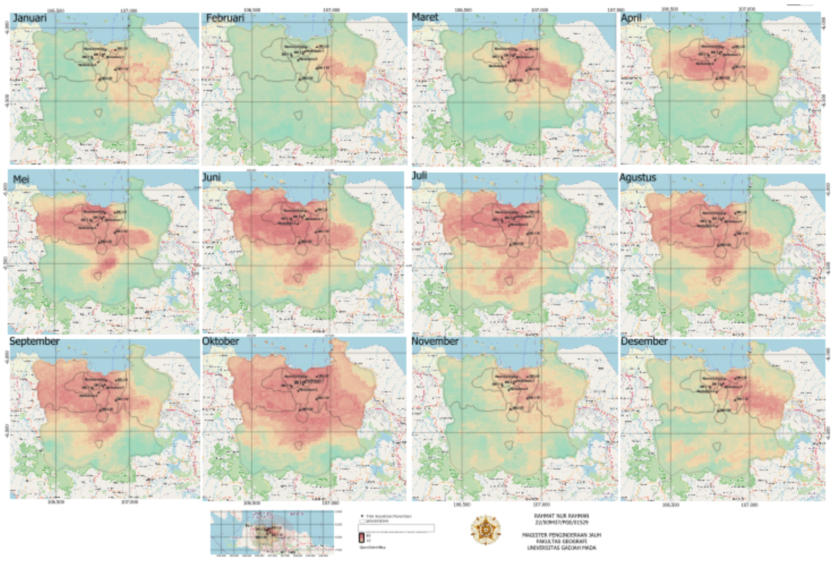


Fig. 9. Distribution Spatial Temporal $PM_{2.5}$ in Jakarta 2021

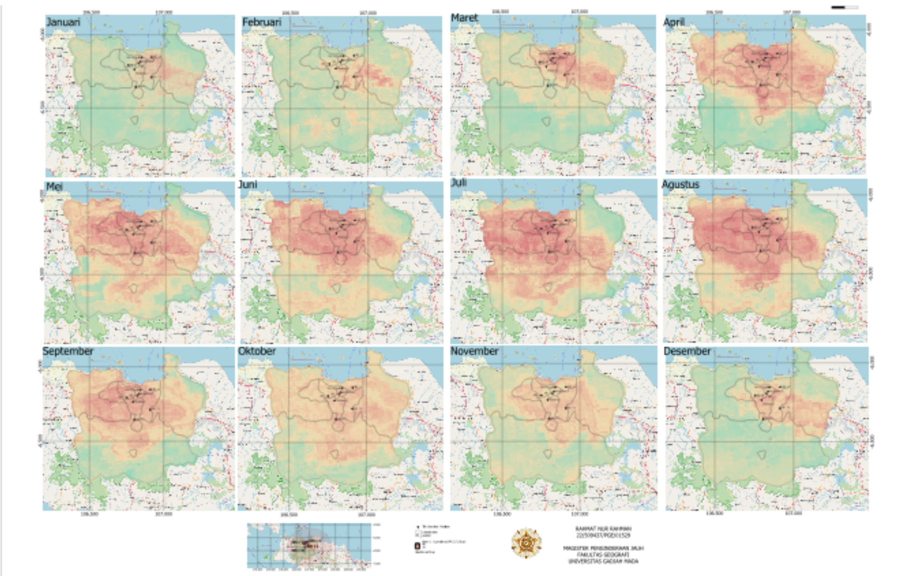


Fig. 10. Distribution Spatial Temporal $PM_{2.5}$ in Jakarta 2022

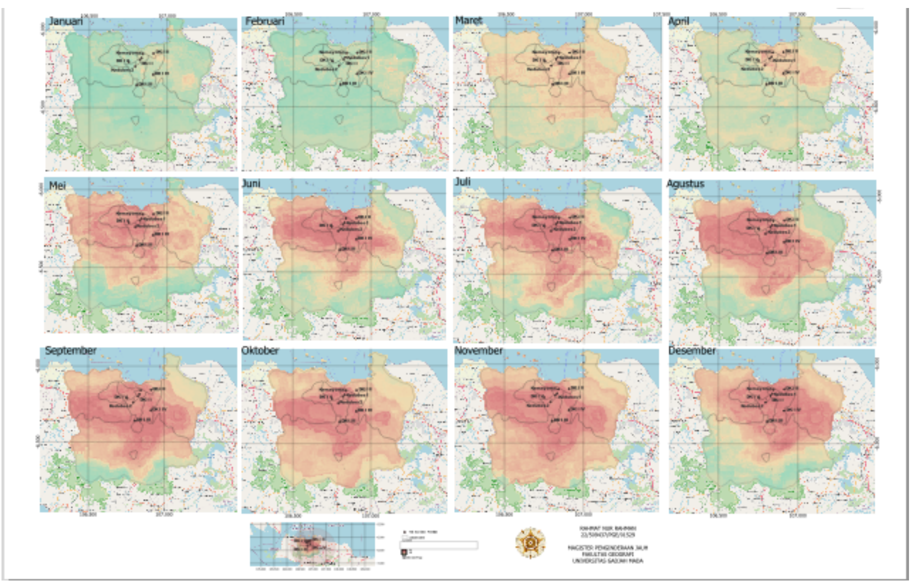


Fig. 11. Distribution Spatial Temporal $PM_{2.5}$ in Jakarta 2023

4. Conclusion

The most influential variable on the $PM_{2.5}$ parameter is NO_2 (Nitrogen Dioxide) and CO (Carbon Monoxide), with the highest correlation value of 0.29. followed by temperature with a correlation of 0.19. Meanwhile, the strongest inverse or negative correlation is found in the wind speed variable, with a correlation value of -0.38.

This study concluded that the best random forest model is when the configuration of trees is but has a value of r^2 = 0.793 and RMSE = 8.28. The spatio-temporal pattern of $PM_{2.5}$ concentration in DKI Jakarta Province shows that the concentration values are quite evenly distributed with relatively high levels, which should be a special consideration in determining future air quality management policies. The spatial pattern formed in this spatial modeling follows the rainy and dry seasons, where the highest spatial pattern values of the $PM_{2.5}$ parameter are observed in the JJA months (June, July, and August), starting to decrease in the SON months (September, October, and November), and finally reaching the lowest values in the DJF months (December, January, and February).

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